

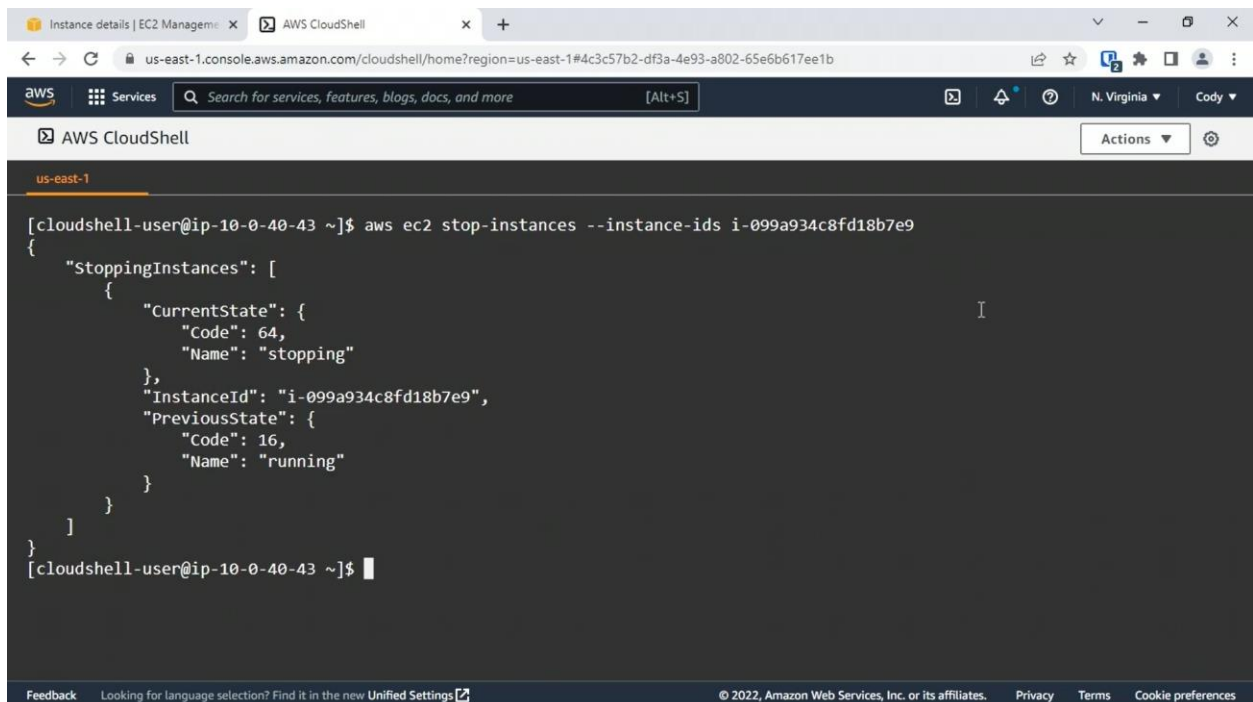
4.4 AWS Resource Scalability

In this course, you will:

- Discover how to resize EC2 instances using the console and the CLI, followed by configuring auto-scaling to support an application workload
- Review when Amazon Fargate, EMR, and Batch should be used
- Discuss when various configurations of ElastiCache should be used
- Review AWS data analytics and ingestion services such as Amazon Athena and AWS Glue

Changing the EC2 Instance Sizing Using the Console

Changing the EC2 Instance Sizing Using the CLI



```
us-east-1
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 stop-instances --instance-ids i-099a934c8fd18b7e9
{
  "StoppingInstances": [
    {
      "CurrentState": {
        "Code": 64,
        "Name": "stopping"
      },
      "InstanceId": "i-099a934c8fd18b7e9",
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
[cloudshell-user@ip-10-0-40-43 ~]$
```

Instance details | EC2 Management | AWS CloudShell

us-east-1.console.aws.amazon.com/cloudshell/home?region=us-east-1#4c3c57b2-df3a-4e93-a802-65e6b617ee1b

aws Services Search for services, features, blogs, docs, and more [Alt+S] N. Virginia Cody

AWS CloudShell Actions

us-east-1

```
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 help
EC2()

NAME
    ec2 -

DESCRIPTION
    Amazon Elastic Compute Cloud (Amazon EC2) provides secure and resizable
    computing capacity in the Amazon Web Services Cloud. Using Amazon EC2
    eliminates the need to invest in hardware up front, so you can develop
    and deploy applications faster. Amazon Virtual Private Cloud (Amazon
    VPC) enables you to provision a logically isolated section of the Ama-
    zon Web Services Cloud where you can launch Amazon Web Services
    resources in a virtual network that you've defined. Amazon Elastic
    Block Store (Amazon EBS) provides block level storage volumes for use
    with EC2 instances. EBS volumes are highly available and reliable stor-
    age volumes that can be attached to any running instance and used like
    a hard drive.
```

it would help me along with the next level commands after aws ec2.

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Instance details | EC2 Management | AWS CloudShell

us-east-1.console.aws.amazon.com/cloudshell/home?region=us-east-1#4c3c57b2-df3a-4e93-a802-65e6b617ee1b

aws Services Search for services, features, blogs, docs, and more [Alt+S] N. Virginia Cody

AWS CloudShell Actions

us-east-1

```
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-attribute --instance-id i-099a934c8fd18b7e9 --attribute instan
ceType
{
  "InstanceId": "i-099a934c8fd18b7e9",
  "InstanceType": {
    "Value": "t2.small"
  }
}
[cloudshell-user@ip-10-0-40-43 ~]$
```

This is exactly what we want.

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The screenshot shows the AWS CloudShell interface in a web browser. The browser's address bar displays the URL: `us-east-1.console.aws.amazon.com/cloudshell/home?region=us-east-1#4c3c57b2-df3a-4e93-a802-65e6b617ee1b`. The AWS CloudShell header includes the AWS logo, a search bar, and navigation links. The terminal window, titled "us-east-1", shows a command prompt `[cloudshell-user@ip-10-0-40-43 ~]$` followed by the command `aws ec2 describe-instance-types --instance-types t2.micro`. A text overlay at the bottom of the terminal reads: "I'm interested in t2.micro, let's say, so I'll put that in."

```
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-types --instance-types t2.micro
```

I'm interested in t2.micro, let's say, so I'll put that in.

The screenshot shows the AWS CloudShell interface. The terminal window shows a command prompt `[cloudshell-user@ip-10-0-40-43 ~]$` followed by the command `aws ec2 describe-instance-types --filters name=hibernation-supported,values=true --query InstanceTypes[].InstanceType`. Below the command, an error message is displayed: "Parameter validation failed: Unknown parameter in Filters[0]: 'name', must be one of: Name, Values; Unknown parameter in Filters[0]: 'values', must be one of: Name, Values". The user then corrects the command to `aws ec2 describe-instance-types --filters Name=hibernation-supported,Values=true --query InstanceTypes[].InstanceType`. A text overlay at the bottom of the terminal reads: "No problem, that's easy to correct."

```
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-types --filters name=hibernation-supported,values=true --query InstanceTypes[].InstanceType
```

Parameter validation failed:
Unknown parameter in Filters[0]: "name", must be one of: Name, Values
Unknown parameter in Filters[0]: "values", must be one of: Name, Values
[cloudshell-user@ip-10-0-40-43 ~]\$ aws ec2 describe-instance-types --filters Name=hibernation-supported,Values=true --query InstanceTypes[].InstanceType

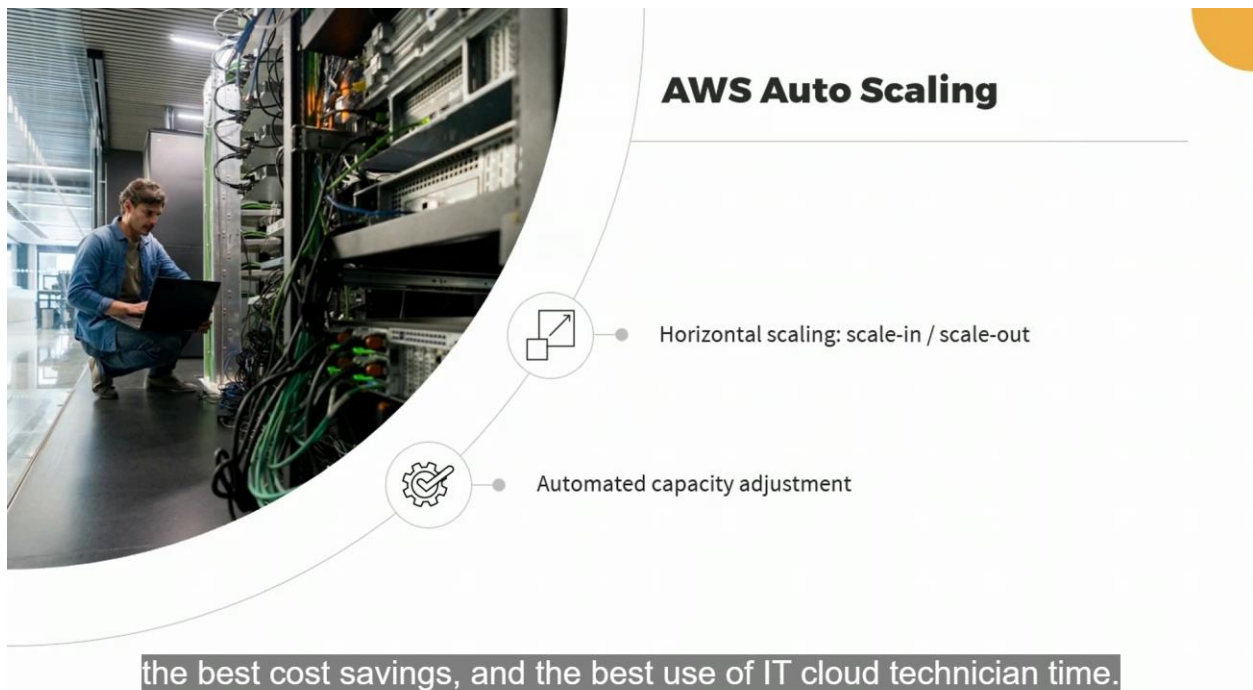
No problem, that's easy to correct.

The screenshot shows the AWS CloudShell interface. The terminal window displays the following commands and output:

```
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 modify-instance-attribute --instance-id i-099a934c8fd18b7e9 --instance-type "{\"Value\": \"t2.micro\"}"
[cloudshell-user@ip-10-0-40-43 ~]$ aws ec2 describe-instance-attribute --instance-id i-099a934c8fd18b7e9 --attribute instanceType
{
  "InstanceId": "i-099a934c8fd18b7e9",
  "InstanceType": {
    "Value": "t2.micro"
  }
}
```

Notice now it's showing a Value of t2.micro.

Auto Scaling



AWS Auto Scaling: Benefits

App performance maintenance during busy peaks



Scaling-in when app usage lessens, which costs less



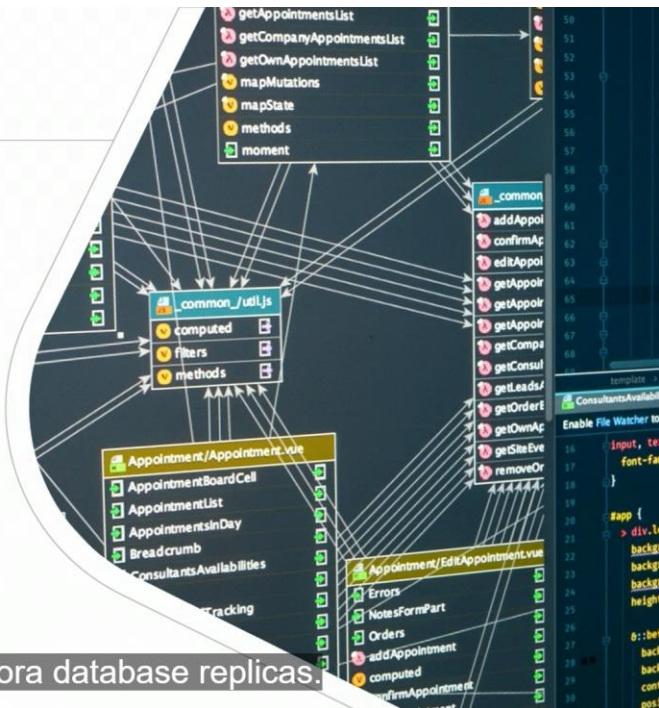
Use simple configuration and automated monitoring to help assess future capacity needs



You can monitor usage over time.

AWS Auto Scaling

- EC2 instances
- Elastic Container Service (ECS)
- Amazon DynamoDB
- Amazon Aurora database replicas



and Amazon Aurora database replicas.

Predictive Scaling



based on past historical activity patterns.

AWS Auto Scaling Plan

AWS Auto Scaling > Scaling plans > Create scaling plan

Step 1
Find scalable resources

Step 2
Specify scaling strategy

Step 3
Configure advanced settings (optional)

Step 4
Review and create

Find scalable resources

Automatically discover or manually choose resources to add to your scaling plan. [Info](#)

Choose a method

☐ **Search by CloudFormation stack**
Search for resources provisioned by an AWS CloudFormation stack.

☐ **Search by tag**
Search for resources by tags applied to them.

☐ **Choose EC2 Auto Scaling groups**
Choose one or more Auto Scaling groups to include in your scaling plan.

we have an example of configuring an AWS Auto Scaling Plan.

AWS Auto Scaling Custom Plan

Scaling strategy
The strategy defines the scaling metric and target value used to scale your resources.

☐ **Optimize for availability**
Keep the average CPU utilization of your Auto Scaling groups at 40% to provide high availability and ensure capacity to absorb spikes in demand.

☐ **Balance availability and cost**
Keep the average CPU utilization of your Auto Scaling groups at 50% to provide optimal availability and reduce costs.

☐ **Optimize for cost**
Keep the average CPU utilization of your Auto Scaling groups at 70% to ensure lower costs.

☒ **Custom**
Choose your own scaling metric, target value, and other settings.

☒ **Enable predictive scaling**
Support your scaling strategy by continually forecasting load and proactively scheduling capacity ahead of when you need it. [Info](#)

☒ **Enable dynamic scaling**
Support your scaling strategy by creating target tracking scaling policies to monitor your scaling metric and increase or decrease capacity as you need it. [Info](#)

▼ **Configuration details**

Scaling metric
Monitored metric that determines if resource utilization is too low or high. [Info](#)

Average CPU utilization

Target value
40 %
Value must be greater than 0 and less than or equal to 100.

Load metric

Average CPU Utilization (Percent)
All scalable targets

100
80.0
60.0
40.0

40%

that's what's been selected, notice the checkbox

Enabling Auto Scaling

AWS Serverless Compute

Amazon Serverless Compute



Managed services (PaaS)



Automates instance deployment and management

like Amazon Fargate, Elastic MapReduce, Amazon Batch.



Amazon Batch

Batch job submission – data analytics, transformation, etc.

Compute and memory power – adjusted manually or automatically

Jobs – scheduled to run on EC2, Spot Instances, or Amazon Fargate

And you schedule the job to run on EC2 instances,

AWS Batch: Job Definition

The screenshot shows the AWS Batch console interface for defining a new job. On the left is a navigation sidebar with options like Dashboard, Jobs, Job definitions, Job queues, Compute environments, Scheduling policies, Wizard, Console settings, and Related AWS services. The main panel is titled 'AWS Batch' and has two tabs: 'Bash' (selected) and 'JSON'. The 'Bash' tab contains a text box for the 'Command' set to 'echo 'hello world'', a section for 'Result of command conversion to JSON' showing '["echo","hello world"]', and dropdown menus for 'vCPUs - required' (set to 1.0) and 'Memory - required' (set to 2 GB). The 'JSON' tab is also visible, describing its use for more complex commands.

You can create a Bash job.

Application Containers



App code and dependencies are packaged into a container



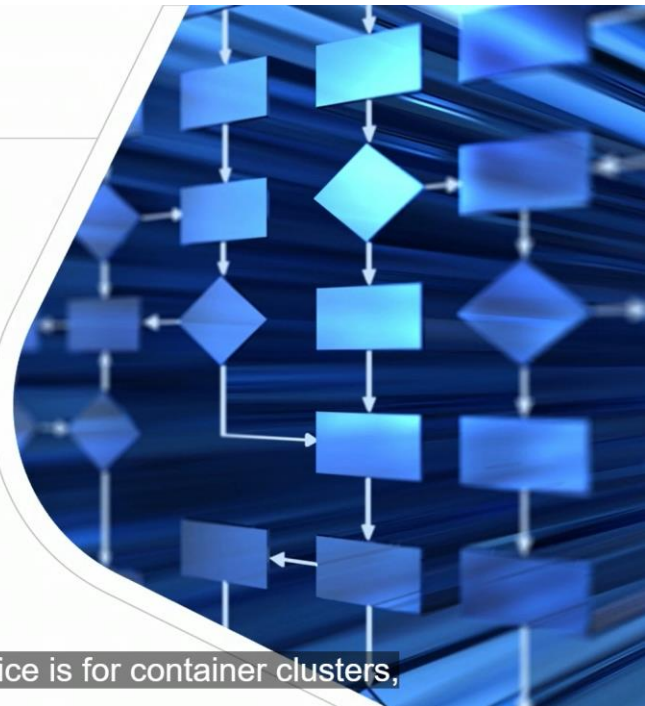
An entire app or app microservice runs in the container



You could have an entire app in a container,

Amazon Fargate

- A serverless compute engine for app containers
- No need to manually manage underlying compute
- Auto scaling – sets resource power to meet needs
- Works with Elastic Kubernetes Service (EKS)
- Works with Elastic Container Service (ECS)



Elastic Kubernetes Service is for container clusters,

Amazon Elastic MapReduce (EMR)

Managed cluster platform that allows distributed processing



Big data analysis



EMR on EC2, EMR on EKS



Based on Apache Hadoop

MapReduce is based on Apache Hadoop.

Amazon ElastiCache

Amazon ElastiCache

Database in-memory caching service – data types/high availability/replication



we have another endpoint available if we've configured it as such.

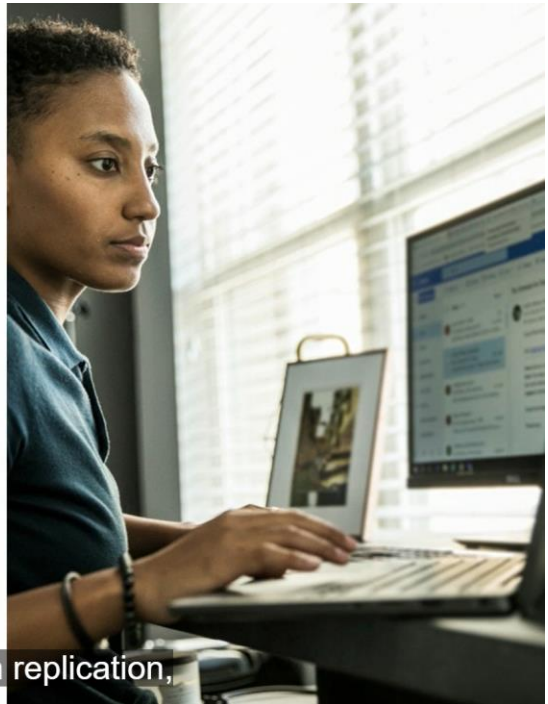
Amazon ElastiCache Engine Options



you get to select the engine option and also the version of Memcached

Memcached

-  Simple key-value storage
-  No built-in replication



Also, there's no built-in replication,

Memcached Cluster Settings

Cluster settings
Use the following options to configure the cluster.

Engine version
Version compatibility of the Memcached engine that will run on your nodes.
1.6.12

Port
The port number that nodes accept connections on.
11211

Parameter groups
Parameter groups control the runtime properties of your nodes and clusters.
default.memcached1.6

Node type
The type of node to be deployed and its associated memory size.
cache.r6g.large
13.07 GiB memory Up to 10 Gigabit network performance

Number of nodes
The number of nodes in this cache cluster. A node is a partition of your data.
1

we have a screenshot of deploying a Memcached cluster.

Redis

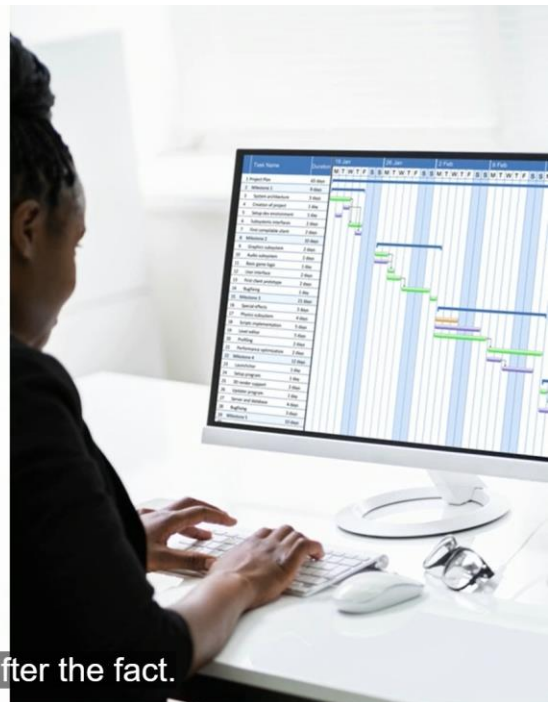


Complex data types beyond value pairs (e.g., hashes, lists, bitmaps, etc.)



Built-in replication, node failover, backup, and restore

You can change that after the fact.



Redis Cluster Settings

Cluster settings
Use the following options to configure the cluster.

Engine version
Version compatibility of the Redis engine that will run on your nodes.
6.2

Port
The port number that nodes accept connections on.
6379

Parameter groups
Parameter groups control the runtime properties of your nodes and clusters.
default.redis6.x

Node type
The type of node to be deployed and its associated memory size.
cache.r6g.large
13.07 GiB memory Up to 10 Gigabit network performance

Number of replicas
Enter the number of replicas between 0 and 5. Zero replicas will not enable an enhanced cluster with primary/replica roles.
2

of deploying ElastiCache using the Redis engine.

AWS Data Analytics and Ingestion Services

Amazon Data Ingestion and Analytic Services

- Big data – introduces processing challenges
- Amount of data
- Rate of data ingestion
- Analysis and resultant insights – compute power required
- Short and long-term data retention



or maybe for regulatory compliance for the long term.

Amazon Athena

- Big data ANSI SQL querying service for data stored in S3
- Managed service
- Data – can be structured, semi-structured, or structureless
- Queries – executed in parallel across nodes for the utmost speed in query results



So at the exact same time across multiple nodes in a cluster

Amazon QuickSight

A Business Intelligence (BI) solution that results in data visualizations and dashboards



Is useful for forecasting trends and conducting what-if scenarios



It supports natural language queries against data from many different sources



And it supports standard querying against many different data sources.

Amazon Kinesis



Used to analyze and process streaming data



Processes data in real time instead of after collection



Is a managed service supporting automatic scaling

You don't have to deploy anything manually.

AWS Glue

Is a serverless extract, transform, load (ETL) service



Consists of a crawler, an ETL service, and a metadata data catalog



Can integrate with Amazon Athena



And it's one of those services that can integrate with Amazon Athena.

AWS Lake Formation Services

- Data lakes – central, large-scale, secure data repositories storing data in its original and transformed-for-analysis form
- Define data sources and security policies
- Seamless integration with other AWS analysis services, such as Amazon Athena



of which large scale data ingestion

1. Which ElastiCache engine is best suited to handle complex data types and replication?
 - Redis
2. To which cloud computing characteristic does auto scaling best apply?
 - Rapid elasticity
3. Which primary element is defined within an EC2 launch configuration?
 - AMI
4. You attempt to change the instance type for a Linux EC2 instance using the console, but the option is grayed out. What is the problem?
 - The instance is running

5. Which CLI command shows the current instance type for an EC2 instance?
 - `aws ec2 describe-instance-attribute`

6. You would like a managed solution that handles the underlying compute sizing for an EKS cluster. What should you use?
 - Fargate

7. Which Amazon service is designed as Business Intelligence (BI) solution?
 - QuickSight